APC 523 Problem Set 4 Solutions

Write-up

April 22, 2026

Problem 1: Elliptic BVP

(a) Reformulate the BVP as a root-finding problem for Newton-Raphson.

Given the non-linear elliptic BVP:

$$\nabla^2 u - u^4 = 0 \quad \text{for } x \in \Omega = [0,1]^2 \tag{1}$$

with $u(x) = 1$ on the boundary $\partial\Omega$.

Discretize the domain into an $N \times N$ regular grid. Using the standard 5-point finite difference stencil for the Laplacian, define the residual function $F(U)$ at each internal node (i,j) as:

$$F_{i,j}(U) = \frac{u_{i-1,j} + u_{i,j-1} - 4u_{i,j} + u_{i+1,j} + u_{i,j+1}}{h^2} - u_{i,j}^4 = 0 \tag{2}$$

where $h = 1/N$. This forms a system of non-linear equations $F(U) = 0$, which is exactly a root-finding problem. To apply the Newton-Raphson method, we require the Jacobian matrix $J(U) = \frac{\partial F}{\partial U}$. The entries of the Jacobian are:

$$\frac{\partial F_{i,j}}{\partial u_{k,l}} = \begin{cases} -\frac{4}{h^2} - 4u_{i,j}^3 & \text{if } (k,l) = (i,j) \text{ (diagonal)} \\ \frac{1}{h^2} & \text{if } (k,l) \text{ are adjacent neighbors of } (i,j) \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

The Newton-Raphson iteration is then defined by solving the linear system:

$$J(U^{(k)})\delta U^{(k)} = -F(U^{(k)}) \tag{4}$$

and updating the state $U^{(k+1)} = U^{(k)} + \delta U^{(k)}$ until $||\delta U^{(k)}||$ or $||F(U^{(k)})||$ is sufficiently small.

Problem 2: Derivation of the Beam-Warming Scheme

To derive the Beam-Warming scheme, we start with the 1D linear advection equation:

$$u_t + au_x = 0 \tag{5}$$

First, we perform a Taylor series expansion of the solution $u(t,x)$ forward in time from t_n to t_{n+1}:

$$u(t_{n+1}, x_i) = u(t_n, x_i) + \Delta t u_t + \frac{\Delta t^2}{2} u_{tt} + O(\Delta t^3) \tag{6}$$

Next, we use the governing PDE to replace the temporal derivatives with spatial derivatives. From the PDE, we know $u_t = -au_x$. Differentiating this relation with respect to time gives us the second derivative:

$$u_{tt} = \frac{\partial}{\partial t}(-au_x) = -a\frac{\partial}{\partial x}(u_t) = -a\frac{\partial}{\partial x}(-au_x) = a^2 u_{xx} \tag{7}$$

Substituting these spatial derivatives back into our Taylor expansion yields:

$$u_i^{n+1} = u_i^n - a\Delta t u_x + \frac{a^2 \Delta t^2}{2} u_{xx} + O(\Delta t^3) \tag{8}$$

To obtain a fully discrete scheme that is second-order accurate in space and utilizes upstream information (assuming $a > 0$), we approximate the continuous spatial derivatives using second-order backward (upwind) finite differences.

For the first spatial derivative:

$$u_x \approx \frac{3u_i^n - 4u_{i-1}^n + u_{i-2}^n}{2h} \tag{9}$$

For the second spatial derivative:

$$u_{xx} \approx \frac{u_i^n - 2u_{i-1}^n + u_{i-2}^n}{h^2} \tag{10}$$

Substituting these discrete approximations into the time-expanded equation, we drop the higher-order error terms to get the numerical approximation:

$$u_i^{n+1} = u_i^n - a\Delta t \left(\frac{3u_i^n - 4u_{i-1}^n + u_{i-2}^n}{2h}\right) + \frac{a^2\Delta t^2}{2}\left(\frac{u_i^n - 2u_{i-1}^n + u_{i-2}^n}{h^2}\right) \tag{11}$$

Finally, rearranging the terms by moving the first derivative approximation to the left side and dividing the entire equation by Δt yields the standard form of the Beam-Warming scheme:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + a\frac{3u_i^n - 4u_{i-1}^n + u_{i-2}^n}{2h} = \frac{a^2\Delta t}{2}\frac{u_i^n - 2u_{i-1}^n + u_{i-2}^n}{h^2} \tag{12}$$

Problem 3: Advection

(a) Analytical Solution

Given the 2D advection equation $u_t + au_x + bu_y = 0$. Let us check the proposed solution $u(t, x, y) = u(0, x - at, y - bt)$. By applying the chain rule with arguments $\xi = x - at$ and $\eta = y - bt$:

$$u_t = -au_\xi - bu_\eta \tag{13}$$

$$u_x = u_\xi \tag{14}$$

$$u_y = u_\eta \tag{15}$$

Substituting into the PDE:

$$(-au_\xi - bu_\eta) + a(u_\xi) + b(u_\eta) = 0 \tag{16}$$

Thus, the analytical solution simply translates the initial condition at speed a in the x-direction and b in the y-direction.

(b)(1) 1D CTU Derivation

Geometrically, consider a piecewise constant function over cells of width Δx. Over one time step Δt, the exact solution shifts the profile to the right by a distance $a\Delta t$. If we average the shifted profile over the original grid cell i, the new cell average u_i^{n+1} is composed of two parts: 1. The portion of cell i's original mass that did not exit the cell: fraction is $(1 - a\Delta t/\Delta x) = (1 - \mu)$. 2. The portion of mass that entered from the upwind cell $i - 1$: fraction is $(a\Delta t/\Delta x) = \mu$. Thus, $u_i^{n+1} = (1 - \mu)u_i^n + \mu u_{i-1}^n$.

(b)(2) 2D CTU Derivation

Extending the geometric approach to 2D, we project the cell (i, j) backward along the characteristic trajectory by distances $(a\Delta t, b\Delta t)$. The projected volume overlaps with four adjacent grid cells at time level n. The normalized overlap areas are: - Overlap with cell (i, j): $(1 - \mu)(1 - \nu)$ - Overlap with cell $(i - 1, j)$: $\mu(1 - \nu)$ - Overlap with cell $(i, j - 1)$: $(1 - \mu)\nu$ - Overlap with cell $(i - 1, j - 1)$: $\mu\nu$

Integrating the piecewise constant values over these overlap areas gives the fully discrete CTU scheme:

$$u_{i,j}^{n+1} = (1 - \mu)(1 - \nu)u_{i,j}^n + \mu(1 - \nu)u_{i-1,j}^n + (1 - \mu)\nu u_{i,j-1}^n + \mu\nu u_{i-1,j-1}^n \tag{17}$$

(b)(3) Modified Equation for CTU

Expanding the 2D scheme using a 2D Taylor series, we gather the leading order terms:

$$u_t + au_x + bu_y = \frac{a\Delta x}{2}(1 - \mu)u_{xx} + \frac{b\Delta y}{2}(1 - \nu)u_{yy} + \dots \tag{18}$$

The CTU scheme eliminates the $O(\Delta t)$ cross-derivative error term (u_{xy}) present in naive dimensional splitting, but remains first-order accurate overall. The leading truncation error consists of numerical diffusion in both the x and y directions.